

Joseth Valdez

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EDUCATION

Bachelor of Science in Computer Science

The University of Texas at El Paso (UTEP)

Expected: May 2027

Overall GPA: 3.56/4.00

Relevant Coursework: Data Structures & Algorithms; Computer Architecture; Operating Systems; Advanced OOP; Automata

Honors and Awards: Honors Program Member – (2023 - Present)

Certifications: Google Cybersecurity Professional Certificate

SKILLS

- **Languages:** Python, Java, C, SQL
- **Libraries:** Pandas, NumPy, Ollama
- **Frameworks:** React, Electron, Vite
- **Technologies/Tools:** Git/GitHub, Linux, Docker, HTTP, CSS, Wireshark, TCPDUMP, Ncat

EXPERIENCE

Cyber Rapid Innovation Group (CyberRig) – The University of Texas at El Paso

El Paso, TX

Undergraduate Researcher

August 2025 – Present

- Automated network data analysis tasks, as evidenced by 3+ Python scripts developed, by processing connection data and extracting patterns from experimental results.
- Simulated 10+ network environments using CORE Network Emulator to test system interactions and evaluate attacker behavior under controlled conditions.
- Led development of cybersecurity research initiatives across 8+ design and testing cycles, coordinating experimental setup and applying findings to measurably improve the system behavior and decoy realism.

Computer Science Department – The University of Texas at El Paso

El Paso, TX

CS IT Support Specialist

January 2025 – February 2025

- Resolved 10+ technical requests per shift spanning hardware, software, and networking issues for faculty, staff, and students.
- Maintained uptime across 10+ CS department workstations by diagnosing recurring issues and applying consistent fixes aligned with security protocols.
- Performed routine maintenance, updates, and backups across CS infrastructure, contributing to consistent system availability throughout the support period.

PROJECTS

Dynamic Honeypot – AI-Driven Adaptive Network Defense System | Bash | Linux | Nmap

- Built 3+ Python scripts to automate network data analysis, parsing connection logs and surfacing behavioral patterns from experimental research configurations.
- Monitored attacker behavior across repeated test scenarios by tracking connection attempts, lateral movement, and service interaction patterns to extract actionable threat intelligence.
- Tuned exposed services and system fingerprints to match realistic infrastructure profiles, reducing early attacker abandonment and improving the quality of observed behavior.

Sentinel Desktop App – AI-Powered Malware Analysis App | React | Electron | Vite

- Built a Windows desktop app (React, Vite, Electron) that scans folders for malicious files by evaluating behavioral context and file patterns, catching threats that static traditional antivirus tools routinely miss.
- Eliminated backend lock-in by abstracting the inference layer to support both local LLM inference (Ollama) and cloud-based models via API, letting users switch environments without any changes to the app.
- Improved threat response time by surfacing LLM-generated explanations of flagged files alongside one-click quarantine, block, or delete options, giving users the context to act immediately instead of guessing.

LEADERSHIP

Cyber Rapid Innovation Group (CyberRIG)

El Paso, TX

Research Contributor & Event Organizer

- Organized a cybersecurity hackathon with 30+ student participants, handling challenge design, logistics, and cross-team coordination from planning through execution.
- Contributed across multiple active research efforts in network security, honeynet design, and adaptive defense strategies.
- Promoted group initiatives and drove student engagement by coordinating research discussions and hands-on activities.